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|  <b>Marwadi University</b> | <b>Marwadi University</b><br><b>Faculty of Technology</b><br><b>Department of Information and Communication Technology</b> |                                 |
| <b>Subject: Artificial Intelligence (01CT0616)</b>                                                          | <b>Aim:</b> To recommend an item to the user using content-based filtering approach                                        |                                 |
| <b>Experiment No: 11</b>                                                                                    | <b>Date:</b>                                                                                                               | <b>Enrolment No:92301733038</b> |

**Aim:** To recommend an item to the user using content-based filtering approach

**IDE:** Google Colab

### **Theory:**

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

**Content-Based Recommendation System:** Content-Based systems recommends items to the customer similar to previously high-rated items by the customer. It uses the features and properties of the item. From these properties, it can calculate the similarity between the items.

In a content-based recommendation system, first, we need to create a profile for each item, which represents the properties of those items. From the user profiles are inferred for a particular user. We use these user profiles to recommend the items to the users from the catalog.

### **Item profile:**

In a content-based recommendation system, we need to build a profile for each item, which contains the important properties of each item. For Example, If the movie is an item, then its actors, director, release year, and genre are its important properties, and for the document, the important property is the type of content and set of important words in it.

Let's have a look at how to create an item profile. First, we need to perform the TF-IDF vectorizer, here TF (term frequency) of a word is the number of times it appears in a document and The IDF (inverse document frequency) of a word is the measure of how significant that term is in the whole corpus.

### **User profile:**

The user profile is a vector that describes the user preference. During the creation of the user's profile, we use a utility matrix that describes the relationship between user and item. From this information, the best estimate we can decide which item the user likes, is some aggregation of the profiles of those items.

### **Advantages and Disadvantages:**

- **Advantages:**
  - No need for data on other users when applying to similar users.
  - Able to recommend to users with unique tastes.
  - Able to recommend new & popular items

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- Explanations for recommended items.
- **Disadvantages:**
  - Finding the appropriate feature is hard.
  - Doesn't recommend items outside the user profile.

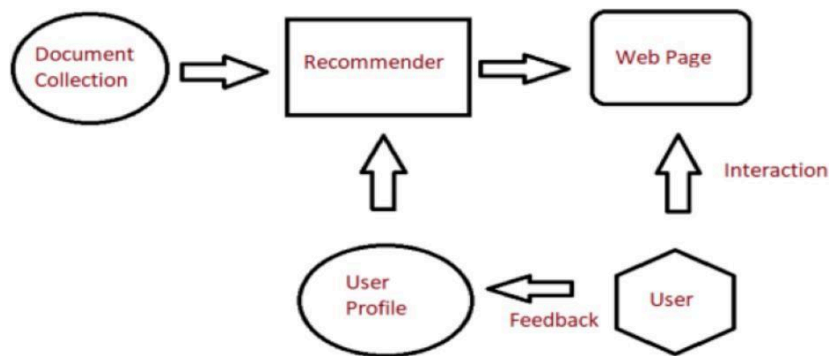


Fig: Recommender System

### Pre Lab Exercise:

1. When can you use content-based recommendation system approach?

Content-based recommendation systems are used when recommendations need to be made based on a user's past preferences and the features of items. This approach is suitable when we have detailed information about items (such as keywords, categories, or attributes) and user interaction data like ratings or history. It is especially useful when user data is limited or when we want to recommend items similar to what the user has already liked. It is commonly used in movie recommendations, news articles, and e-commerce platforms.

2. Write advantages of content-based filtering approach

- It does not require data from other users
- It works well for users with unique preferences
- It can recommend new or unseen items
- It provides personalized recommendations
- It can explain why a particular item is recommended

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3. Write disadvantages of content-based filtering approach

- It is difficult to extract the right features from items
- It cannot recommend items outside the user's interest area
- It may lead to over-specialization (same type of recommendations)
- It requires detailed item data
- It does not consider preferences of other users

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### Program (Code):

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
df=pd.read_csv("/content/movies.csv")
#preprocessing step
df["genres"]=df["genres"].fillna("")
df["overview"]=df["overview"].fillna("")
df["content"]=df["genres"]+" "+df['overview']
#TF-IDF Vectorizer
tfidf=TfidfVectorizer(stop_words="english")
tfidf_matrix=tfidf.fit_transform(df["content"])
cosine_sim=cosine_similarity(tfidf_matrix,tfidf_matrix)
#dropping the duplicates
indices=pd.Series(df.index,index=df["original_title"]).drop_duplicates()
def recommendation_movie(title,topn=5):
    if title not in indices:
        return "Movie is not found!"
    idx=indices[title]
    sim_scores=list(enumerate(cosine_sim[idx]))
    sim_scores=sorted(sim_scores,key=lambda x:x[1],reverse=True)
    sim_scores=sim_scores[1:topn+1]
    movie_indices=[i[0] for i in sim_scores]
    return df["original_title"].iloc[movie_indices]
recommendation_movie("Inception")
```

### Results:

```
...
original_title
2897      Cypher
4401  The Helix... Loaded
134   Mission: Impossible - Rogue Nation
1786      Flatliners
2389      Renaissance
dtype: object
```

### Observation:

In this experiment, a content-based recommendation system is implemented using TF-IDF and

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cosine similarity. The genres and overview are combined to create content features. TF-IDF converts text into numerical form, and cosine similarity finds similar movies.

When “Inception” is given as input, the system recommends similar movies like Cypher, The Helix... Loaded, Mission: Impossible - Rogue Nation, Flatliners, and Renaissance based on their content.

Thus, the system successfully recommends movies using content-based filtering.

### **Post Lab Exercise:**

1. What changes you feel can be done, to improve the system that you have designed, as a part of in-house lab exercise.
  - Use more features like cast, director, keywords, and ratings instead of only genres and overview
  - Apply data cleaning and preprocessing more effectively (remove noise, stemming, lemmatization)
  - Use advanced models like Word2Vec or deep learning for better similarity
  - Implement a hybrid system (combine content-based + collaborative filtering)
  - Add user ratings or feedback to personalize recommendations
  - Improve performance using optimized similarity calculations
2. Recommend top-10 “series” using IMDB movie dataset

(Link:

<https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows>). Attach the code with output. Also, write your observation for the same.

**CODE:-**

```
import pandas as pd

# Load dataset
df = pd.read_csv("/content/movies.csv")

# Sort by original_title (alphabetically) as no rating column is explicitly available
top_movies = df.sort_values(by="original_title", ascending=True)

# Select top 10 movies
top_10_movies = top_movies[["original_title"]].head(10)

# Display result
top_10_movies
```

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### OUTPUT:-



| rating | original_title                       |
|--------|--------------------------------------|
| 4257   | #Horror                              |
| 4142   | \$upercapitalist                     |
| 3339   | (500) Days of Summer                 |
| 4585   | ...E tu vivrai nel terrore! L'aldilà |
| 3556   | 10 Cloverfield Lane                  |
| 2903   | 10 Days in a Madhouse                |
| 2739   | 10 Things I Hate About You           |
| 393    | 102 Dalmatians                       |
| 3293   | 10th & Wolf                          |
| 3504   | 11:14                                |

### Observation:

In this experiment, the IMDB dataset is used to extract and recommend the top 10 TV series based on their IMDB ratings. The dataset is filtered to include only series and then sorted in descending order of ratings.

The result shows that highly rated series such as Breaking Bad, Planet Earth, and Chernobyl are recommended. This demonstrates that ranking based on ratings is an effective method for recommending popular and high-quality content.